



LME49710 Based AUDIO AMPLIFIER

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Listening to music on a self-made system is great. Good-quality audio amplifier circuits can be constructed with minimum components. Described here is an LME49710 based audio amplifier. Author's prototype is shown in Fig. 1.



Fig. 1: Author's prototype

Circuit and working

As shown in Fig. 2, circuit diagram of the LME49710 based power amplifier consists of power supply unit and audio amplifier unit.

The power supply unit is a linear type built around transformer X1, bridge rectifier DB107 (BR1), 12V voltage regulator LM7812 (IC1) and a few other components. The 230V AC primary of X1 is stepped down to 15V, 1A. DB107 converts AC to pulsating DC, filtered with capacitors C1 and C2, and the rectified output is given to 12V regulator

(IC1). Diode 1N4007 (D1) is used to protect IC1 from reverse voltage. LED1 is used as the power indicator. Resistor R1 is used to limit the current passing through LED1.

The audio amplifier circuit is built around LME49710 (IC2), which is a high-performance, high-fidelity, audio-operational amplifier. The 12V power supply is connected to op-amp power pins. The single-supply operation requires a virtual ground at a voltage level of $V_{cc}/2 = 12V/2 = 6V$ at the non-inverting terminal of op-amp IC2.

The circuit can be constructed with a simple voltage divider formed by resistors R4 and R5 and bypassed by capacitor C8. Capacitors C4 and C5 decouple the noise at IC level, where C5 works as an electrolytic decoupling capacitor to reduce low- and mid-frequency noise and C4 reduces high-frequency noise in the power supply lines.

Audio input is connected at connector CON4. Capacitor C9 blocks DC voltage at input stage and potmeter VR1 controls the volume, up and down.

Resistors R7 and R6 set the gain of amplifier IC2. R7 and capacitor C7 limit the bandwidth of the amplifier and keep it linear, resistors R2 and R3 set the bias current of Vbe multiplier formed by diodes D2 and D3, and potmeter VR2 helps with bias current adjustment. Vbe multiplier is the DC bias voltage that overcomes turn-on base-emitter voltage drops of output transistors T1 and T2.

DC bias also keeps T1 and T2 active when no load current is being delivered to the speaker or load. T1 and T2 drive the speaker through capacitor C10, which also blocks DC. Resistor R8 and

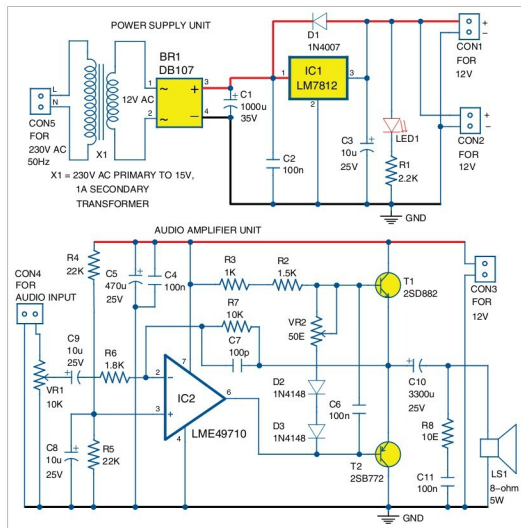


Fig. 2: Circuit diagram of the LME49710 audio amplifier